

MOBRIDGE, SD, USA

WMO: 726685

Lat: 45.546N

Lon: 100.408W

Elev: 1715

StdP: 13.81

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94052

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-8.7	-19.9	2.0	-13.0	-15.2	2.6	-8.1	32.8	19.2	28.4	19.2	6.6	310	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	95.0	70.1	91.0	69.5	87.8	68.7	74.9	86.7	73.0	85.0	71.1	83.2	13.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	125.0	81.5	69.5	115.8	79.5	67.1	106.1	77.6	39.6	86.2	37.7	85.0	36.1	83.5	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.4	25.4	22.8	-22.8	102.1	5.9	5.3	-27.1	106.0	-30.5	109.1	-33.8	112.1	-38.0	116.0		
			-23.1	78.4	5.7	2.2	-27.2	80.0	-30.6	81.3	-33.8	82.5	-37.9	84.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	16.1	19.7	31.6	45.0	56.3	66.8	74.0	71.8	62.6	47.6	33.0	20.0
	DBStd	22.77	13.97	13.71	13.25	10.44	8.58	7.24	6.28	6.32	8.99	9.48	11.21	12.53
	HDD50	4339	1051	847	581	210	38	0	0	0	9	158	514	931
	HDD65	7844	1516	1267	1036	601	291	63	6	14	150	545	959	1396
	CDD50	2701	0	0	10	60	232	505	743	677	386	83	5	0
	CDD65	731	0	0	0	2	21	117	284	226	77	4	0	0
	CDH74	8100	0	0	6	61	327	1260	3078	2389	904	75	0	0
	CDH80	3398	0	0	0	13	93	459	1415	1049	350	19	0	0
Wind	WSAvg	10.5	9.7	9.7	10.9	11.8	11.5	10.4	9.9	10.0	10.7	10.9	10.0	9.7
Precipitation	PrecAvg	17.0	0.4	0.4	0.6	1.3	2.8	3.4	2.3	2.1	1.4	1.4	0.4	0.4
	PrecMax	22.8	1.2	1.6	2.5	4.5	8.2	8.1	5.4	5.8	4.4	5.4	1.3	1.5
	PrecMin	8.1	0.1	0.0	0.1	0.2	0.3	0.8	0.7	0.5	0.0	0.0	0.0	0.0
	PrecStd	3.6	0.2	0.3	0.5	0.9	1.7	1.5	1.2	1.3	1.1	1.3	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	54.8	72.3	81.3	88.1	95.0	102.5	99.5	95.2	82.2	66.4	51.4
		MCWB	40.9	43.1	52.5	57.7	65.7	69.9	71.3	70.5	68.3	59.5	50.6	41.6
	2%	DB	43.0	46.3	63.3	73.4	82.1	89.6	96.3	93.5	88.8	73.7	59.3	45.3
		MCWB	36.9	39.3	49.4	54.1	63.3	69.0	71.4	70.3	67.0	56.9	47.1	38.1
	5%	DB	38.4	42.0	56.3	68.1	77.1	85.0	91.4	89.9	83.3	68.0	53.8	39.9
		MCWB	34.2	36.5	45.3	51.8	60.4	67.3	70.2	69.2	65.2	53.8	43.7	34.4
	10%	DB	35.4	37.3	50.3	63.1	72.6	81.4	87.9	86.0	79.1	62.9	48.2	36.2
		MCWB	32.2	33.4	42.4	48.9	58.3	65.8	69.8	67.7	63.1	51.7	41.2	32.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.2	44.8	54.7	59.8	69.1	76.1	78.3	76.3	72.9	62.2	52.2	42.2
		MCDB	49.6	52.0	67.0	75.9	82.3	87.4	89.9	88.9	84.8	76.7	63.5	49.7
	2%	WB	37.1	39.8	50.3	55.4	65.8	72.6	75.6	73.4	69.8	58.8	47.9	38.3
		MCDB	42.1	46.1	62.3	71.9	78.4	84.4	86.9	86.3	83.8	70.7	58.6	44.8
	5%	WB	34.5	36.8	46.2	52.6	62.4	70.0	73.8	71.4	67.5	55.1	44.3	35.0
		MCDB	38.3	41.6	56.1	65.7	73.4	81.5	85.5	83.6	79.5	66.6	52.5	39.6
	10%	WB	32.0	33.7	42.1	50.1	59.7	67.9	72.0	69.8	65.0	52.5	41.3	32.4
		MCDB	35.2	37.3	50.3	61.6	69.6	78.8	83.9	82.2	76.0	61.4	47.9	36.0
Mean Daily Temperature Range	5% DB	MDBR	18.5	19.0	20.6	23.3	22.8	22.4	23.7	24.5	24.3	21.4	19.5	17.4
		MCDBR	22.5	23.6	31.3	33.2	30.2	27.7	29.5	29.7	30.4	31.0	28.4	23.5
		MCWBR	17.3	16.9	18.7	17.5	15.2	12.6	10.9	11.4	12.7	16.2	17.3	17.1
	5% WB	MCDBR	21.5	22.5	30.5	30.2	26.5	24.0	23.9	25.6	26.2	27.4	27.2	22.9
		MCWBR	16.8	16.4	18.6	16.5	14.7	12.9	11.5	11.8	12.7	15.0	17.1	16.9
Clear-Sky Solar Irradiance	taub		0.258	0.265	0.290	0.326	0.353	0.368	0.376	0.377	0.343	0.307	0.273	0.256
	taud		2.487	2.500	2.508	2.414	2.413	2.405	2.413	2.386	2.463	2.533	2.524	2.499
	Ebn at Noon		278	297	302	297	290	285	281	276	278	276	269	266
	Edh at Noon		21	25	29	35	36	36	36	35	30	24	20	18
All-Sky Solar Radiation	RadAvg		552	885	1207	1569	1861	2108	2160	1807	1414	906	597	439
	RadStd		44	84	104	111	171	143	95	148	95	112	70	24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page